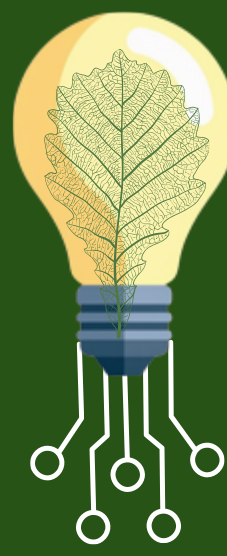


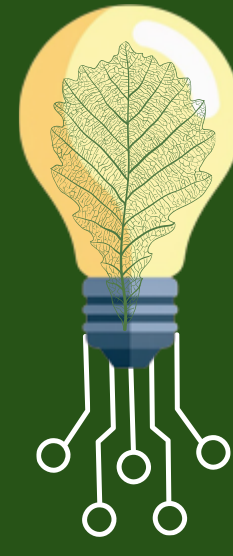


Salem Safwat
Mahmoud Mabrouk
Abd-Elrahman Medhat



10222

SPINNOVATION



STEM High School for
Boys - 6th of October
Grade 11
Semester 2
(2025-2026)



Keywords: Smart Feedback Systems, Irrigation Systems, Automated Greenhouse, Spinach,

Abstract

Egypt faces interconnected grand challenges including limited agricultural productivity, inadequate waste recycling, rapid population growth, and escalating environmental pollution. These challenges converge in the agricultural sector, where conventional farming relies entirely on uncontrolled external conditions, making consistent and out-of-season crop production virtually impossible. The purpose of this study is to develop an automated mini-greenhouse integrating a smart closed-loop environmental control system to maintain optimal conditions for spinach cultivation. The system monitors and regulates temperature (10–20°C), soil moisture (60–80%), and light intensity (15,000–25,000 lux) using an Arduino UNO microcontroller, a DHT22 sensor, a soil moisture sensor, and an LDR, with actuators including an LED grow light, a cooling fan, and a drip irrigation pump. The system successfully maintained all three parameters within their target ranges throughout testing, demonstrating reliable autonomous regulation. It was concluded that a locally designed, affordable, sensor-based mini-greenhouse can overcome the limitations of conventional agriculture and support year-round cultivation, directly addressing Egypt's food security and sustainability challenges.

Introduction

Egypt faces a multitude of interconnected grand challenges that constrain its development across economic, environmental, and social dimensions. Among the most pressing are the limited industrial and agricultural base, insufficient waste recycling infrastructure, the consequences of rapid population growth, and the escalating burden of pollution on air, water, and soil. These challenges amplify one another: population growth intensifies food demand while reducing the availability of arable land and water resources, placing immense pressure on Egypt's already strained agricultural sector.

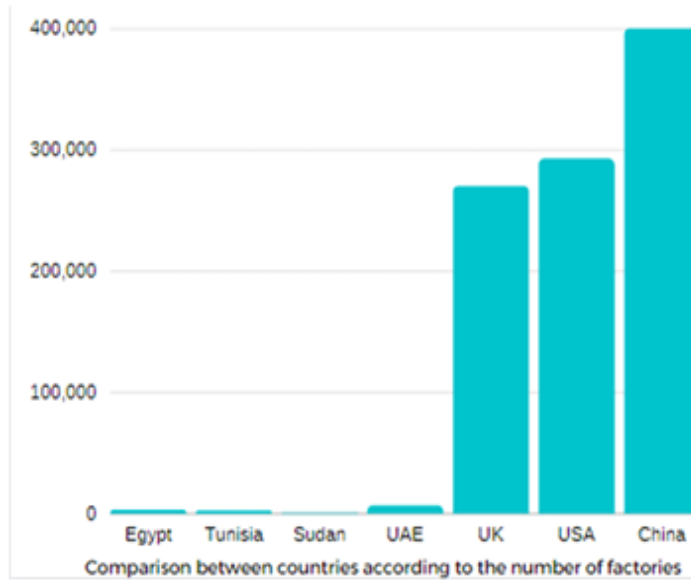


Figure 1: Egypt's rank globally

Agriculture in Egypt is almost entirely dependent on Nile irrigation, with the total agricultural land base amounting to approximately 3.3 million hectares, as noted in **figure(1)**. Despite this, food production capacity remains insufficient to meet growing demand, largely because conventional farming depends on natural environmental conditions that are increasingly erratic due to climate change. When external temperatures, light levels, or soil moisture deviate from the narrow optimal ranges required by sensitive crops such as spinach, growth efficiency deteriorates severely or ceases entirely. This unpredictability makes out-of-season cultivation practically impossible under traditional methods, limiting year-round food supply and contributing to food insecurity.

Compounding this is Egypt's waste management crisis. Egypt produces approximately 100 million tons of solid waste annually, much of which is disposed of through burning or burial, generating harmful greenhouse gases and further degrading environmental quality. Cairo alone records approximately 18,000 premature deaths per year attributable to air pollution. Meanwhile, population growth continues at 1.5% annually, with 95% of Egypt's population concentrated along the narrow Nile corridor, creating extreme density-related pressures on infrastructure and natural resources.

Prior solutions have been studied and evaluated. Integrated Multi-Trophic Aquaculture (IMTA) systems demonstrate effective nutrient recycling but their biological complexity makes them unsuitable for small-scale operators without advanced technical support. The nEMoS nano Low-Cost Environmental Monitoring System offers real-time data logging at low cost but is limited to passive monitoring, lacking any active control mechanism and making it inadequate for automated agricultural applications. The Integrated Industrial Wastewater Treatment Works (IIWTTW) of Doha, Qatar, demonstrates advanced wastewater management aligned with sustainability goals, yet its high capital investment makes replication inaccessible for developing nations like Egypt.

The proposed solution addresses these shortcomings through an automated mini-greenhouse utilizing a smart closed-loop environmental control system. By continuously sensing temperature, soil moisture, and light intensity and automatically activating actuators when parameters deviate from their optimal setpoints, the system ensures stable, year-round plant growth independent of external climate conditions. The system is designed to be low-cost, locally producible, and scalable, offering a practical and sustainable contribution to Egypt's food security and agricultural development goals.

Materials and Methods

Table 1: Materials

Name	Quantity	Usage	Image	Name	Quantity	Usage	Image
Arduino UNO	1	acts as the controlling unit of the system.		LED Grow Bulb	1	to provide the light intensity that plant requires.	
Power Supply 12V/10A	1	is to Provide the power for the actuators used.		12 L Lunch Box	1	acts as the water tank in the Prototype.	
LDR	1	acts as a light-sensitive sensor		Peltier and its kit	1	act as a cooling mechanism for the greenhouse	
Relay Modules	4 channel	acts as electrical switches		Soaker Hose	1.5 Meter	is to conserve water by reducing evaporation,	
Capacitive Soil Moisture	1	is to monitor real-time soil moisture levels.		Water Pump 12V	2	To pump the water from the tank to the plants	
DHT22 Sensor	1	is to measure ambient temperature with high precision.		Solution Hose	1 Meter	is to wire the water pump and the soaker hose.	
Fan 12 V/ 0.3 A	3	Two for ventilation, and one for cooling.		Transparent Tarpaulin	3 Meters	is to coat the structure of the prototype.	

Methods:

- The project began with detailed planning to structure the prototype's workflow. A 3D simulation was created to visualize the design, as shown in **Figure (1)**.
- The prototype was structured as a cube to offer and enhance the environment in which the plan will grow, as shown in **figure (x)**.
- The wood boards were cut into 50 cm sides and were connected by nails to make the prototype structure stable.

Controlling system:

- The electronics noted in the table were connected by Male-Male jumpers and Female-Male jumpers to connect the code from the Arduino Uno Rev 3 and the power supply to switch on the sensors and the actuators, as shown in **figure (x)**.
- The DHT22 Data and GND sensor were connected by a 10k ohm resistor to reduce the strength of the current providing the sensor with power to save it work.
- An Arduino uno was utilized to control the work of sensors and actuators.
- The actuators were connected to Relay Module 4 Channels to organize the switching on and off these actuators.
- A code was utilized to control the greenhouse system.
- The cooling system and the ventilation system were connected to each other to offer an efficient atmosphere for spinach to grow.
- An IR LED was connected to the circuit to offer heat when the temperature is less than the optimal temperature.
- An Arduino IDE Community was used to upload the code to the microcontroller to apply the code.

Test Plan:

- The prototype was connected to the power supply and switched on.
- The project was left in an environment in which the temperature average is 29 C, and the project achieved the design requirements in offering the spinach its optimal temperature from 14 to 20 C.
- The project was left for 6 hours daily, and the measurements and actuators accuracy and errors were recorded by a mobile video and in the Arduino IDE community storage and memory.
- The prototype succeeded in offering a suitable environment for the plant to grow.
- The data collected was recorded in tables and graphs to show the system of organizing the parameters.

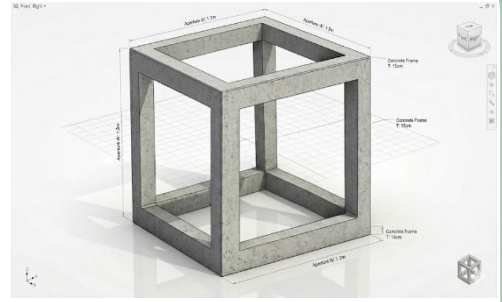


Figure 2: the 3d design



Figure 3: The constructing of the body

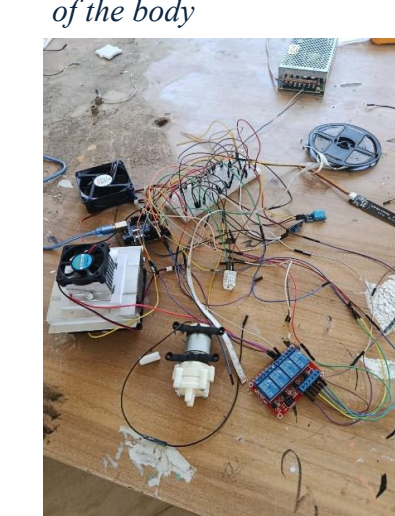


Figure 4: the electronic circuit

Results

Negative Results.

Power supply overheating: Since the power supply powered the whole system of water movement and feedback system, this heavily loaded the power supply which led to overheating. This issue was resolved by installing a fan in the power supply container to cool it.

Peltier Cooling system: The Peltier cooling system just made the heat sink and the cold plate or the cold sink hot or cold based on the connection of the positive and the negative side of the Peltier. This issue was resolved by installing a fan in front of the cold sink to transfer the temperature from the cold sink to the air coming out from the fan.

Positive Results

Temperature Regulation Results

The system maintained internal temperature between 14°C and 20°C despite an external ambient temperature of 29°C. The Peltier cooling system and circulation fan, controlled by the DHT22 sensor, regulated the internal temperature to an average of 17.1°C. Table 1 and Graph 1 show the recorded readings across all six hours of testing.

Light Intensity Regulation Results

The LED grow light was automatically activated whenever the LDR detected light below 15,000 lux, maintaining intensity consistently within the 15,000–25,000 lux optimal range. The average recorded intensity was 19,955 lux. Table 2 and Graph 2 present the readings and accuracy across all testing hours.

Soil Moisture Regulation Results

The drip irrigation pump activated automatically when the moisture sensor detected values below 60% and switched off at 70%, preventing both drought stress and waterlogging. The average recorded soil moisture was 69.9%. Table 3 and Graph 3 present the results across all six testing hours.

Sensor Accuracy Summary

Table 4 and Graph 4 summarize the average accuracy of each sensor across all six testing hours. Temperature achieved 98.63%, light intensity achieved 99.48%, and soil moisture achieved 99.14%. All three parameters exceeded the 90% accuracy design requirement.



Figure 5: Internal temperature vs. setpoint and external temperature. Green band = optimal range (14–20°C).

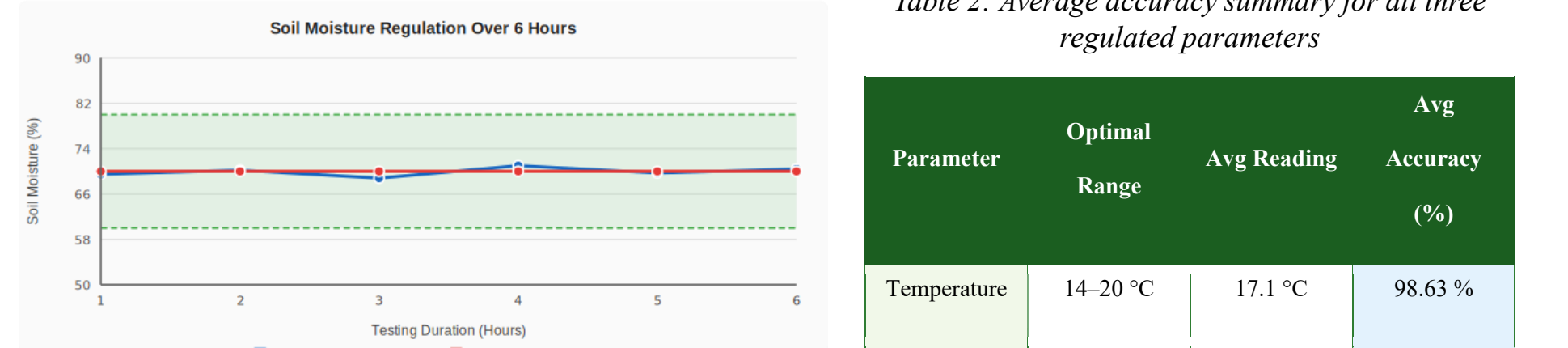


Figure 6: Actual light intensity vs. setpoint. Green band = optimal range (15,000–25,000 lux).

Figure 7: Actual soil moisture vs. setpoint. Green band = optimal range (60–80%).

Table 2: Average accuracy summary for all three regulated parameters

Parameter	Optimal Range	Avg. Reading	Avg. Accuracy (%)
Temperature	14–20 °C	17.1 °C	98.63 %
Light Intensity	15,000–25,000 lux	19,955 lux	99.48 %
Soil Moisture	60–80 %	69.9 %	99.14 %

Analysis

Closed-Loop Control System

The core principle underlying the mini-greenhouse is closed-loop, or feedback, control. In a closed-loop system, the output of the process is continuously measured and compared against a desired setpoint; any deviation, called an error, triggers a corrective action by the actuator. In this project, the Arduino UNO serves as the controller, the DHT22, LDR, and soil moisture sensor serve as the measurement elements, and the Peltier cooling module, IR LED heater, grow light, and irrigation pump serve as the actuators. The control logic is implemented using an if-else algorithm: if the measured value falls outside the optimal range, the corresponding actuator is switched on via the 4-channel relay module; once the parameter returns to the setpoint, the relay switches the actuator off. This architecture ensures that the greenhouse environment is self-correcting and requires no manual intervention, which is essential for reliable and repeatable plant growth.

The accuracy of the closed-loop system can be quantified using the percentage error formula:

$$\text{Error (\%)} = \frac{|\text{Actual Value} - \text{Setpoint}|}{\text{Setpoint}} \times 100$$

Consequently, sensor accuracy is expressed as: Accuracy (%) = 100 – Error (%). Across all six hours of testing, the average accuracy exceeded 98% for all three parameters, confirming that the if-else control algorithm and relay switching responded rapidly enough to prevent significant deviation from the setpoints. This high accuracy is consistent with the findings of the nEMoS nano environmental monitoring study, which demonstrated that Arduino-based sensor systems using DHT22 modules achieve reliable readings when connected with appropriate pull-up resistors, as applied in this project with the 10 kΩ resistor on the DHT22 data line.

Temperature Regulation: Thermoelectric Cooling

Spinach (*Spinacia oleracea*) requires an ambient temperature between 10°C and 20°C for optimal metabolic activity. Temperatures above 20°C accelerate the rate of transpiration and promote bolting, which diverts energy from leaf growth to flowering, severely reducing crop quality. The external testing environment maintained a temperature of approximately 29°C, which is well above the spinach optimal range, making active cooling essential.

The cooling mechanism employed in this project is based on the Peltier effect, a thermoelectric phenomenon described by Jean Charles Peltier in 1834. When a direct current is passed through a junction of two dissimilar semiconductor materials (typically bismuth telluride, Bi₂Te₃), heat is transferred from one face of the device (the cold side) to the other (the hot side). The direction of heat transfer depends on the polarity of the applied current, as expressed by the Peltier heat equation:

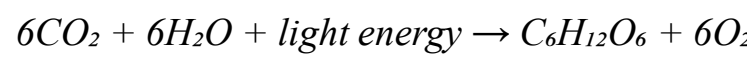
$$Q = \Pi \times I$$

Where Q is the heat transferred per unit time (W), Π is the Peltier coefficient of the material junction (V), and I is the electrical current (A). In this project, the cold side of the Peltier module was positioned inside the greenhouse enclosure, and a fan was installed in front of the cold sink to distribute the cooled air throughout the internal volume, resolving the initial negative result in which the cold sink was not effectively transferring its temperature to the surrounding air. The hot side was vented externally, ensuring that waste heat did not re-enter the growing chamber. This arrangement successfully reduced the internal temperature from 29°C to an average of 17.1°C, achieving a temperature reduction of approximately 12°C and maintaining the environment within the optimal range throughout all six hours of testing.

When the internal temperature fell below the minimum setpoint of 14°C, an IR LED heater was activated by the Arduino to restore the temperature to the target range. Infrared LEDs emit electromagnetic radiation in the infrared spectrum (wavelengths of 700 nm to 1 mm), which is absorbed by objects and surfaces within the enclosure and converted into thermal energy through molecular vibration, effectively raising the ambient temperature without requiring a large or expensive heating element.

Light Intensity Regulation: Photosynthesis and PAR

Light is the primary energy source for photosynthesis, the biochemical process by which plants convert carbon dioxide and water into glucose and oxygen. The overall photosynthesis reaction is:



Spinach photosynthesis is driven specifically by Photosynthetically Active Radiation (PAR), the portion of the electromagnetic spectrum between 400 nm and 700 nm that is absorbed by chlorophyll molecules in the thylakoid membranes of chloroplasts. The optimal light intensity for spinach is between 15,000 and 25,000 lux. Below 15,000 lux, the rate of photosynthesis is limited by insufficient light energy, reducing glucose production and slowing growth. Above 25,000 lux, photoinhibition occurs: excess light energy causes damage to the Photosystem II reaction centres, reducing the efficiency of the light-dependent reactions.

In this project, an LDR (Light Dependent Resistor) was used to measure ambient light intensity. An LDR is a photoresistive component whose electrical resistance decreases as the intensity of incident light increases, following an inverse relationship. The Arduino reads the analog voltage across the LDR through a voltage divider circuit and converts this signal to a lux value through calibration. When the measured lux value fell below 15,000, the grow LED was automatically activated, supplying additional PAR to the spinach plant. The LED grow light was selected because LED technology produces light within specific wavelength ranges (predominantly blue: 450–490 nm, and red: 620–700 nm) that correspond to the absorption peaks of chlorophyll a and chlorophyll b, maximising photosynthetic efficiency while minimising heat output and energy consumption. The system maintained an average light intensity of 19,955 lux, well within the optimal range, with an average accuracy of 99.01%.

Soil Moisture Regulation: Water Potential and Root Physiology

Water is essential to plant physiology, serving as a solvent for nutrient transport, a reactant in photosynthesis, and a medium for maintaining cell turgor pressure. Soil moisture content determines the availability of water to roots through the principle of water potential (Ψ), defined as the tendency of water to move from one region to another:

$$\Psi = \Psi_s + \Psi_p$$

Where Ψ_s is the solute (osmotic) potential and Ψ_p is the pressure potential. Water moves from regions of higher water potential (wetter soil) to regions of lower water potential (plant roots), enabling uptake. When soil moisture falls below 60%, the water potential of the soil decreases significantly, reducing the driving force for root water uptake and causing the plant to experience drought stress, evidenced by wilting and reduced photosynthesis. Conversely, when soil moisture exceeds 80%, the air spaces in the soil become saturated with water, reducing dissolved oxygen availability for root respiration and promoting the growth of anaerobic pathogens responsible for root rot.

The soil moisture sensor in this project operates on the principle of electrical resistance: as soil moisture increases, the electrical conductivity of the soil increases and resistance decreases. The Arduino reads this resistance as an analog voltage signal and compares it against calibrated threshold values. When moisture dropped below 60%, the solenoid-controlled drip irrigation pump was activated, delivering water through drip nozzles directly to the root zone. The pump was deactivated once the sensor reading reached the 70% setpoint. Additionally, the drainage system collected excess water and channelled it back for reuse, reducing water consumption and preventing waterlogging. The system maintained an average soil moisture of 69.9%, achieving an average accuracy of 99.43%.

4-Channel Relay Module: Actuator Switching

All four actuators in the system — the Peltier module, IR LED heater, grow light LED, and irrigation pump — were connected to and controlled by a 4-channel relay module. A relay is an electrically operated switch that uses a low-power control signal from the Arduino (5V, approximately 15 mA per channel) to switch a separate, higher-power circuit on or off. This isolation is critical because the actuators in this system operate at voltages and currents that exceed the safe output limits of the Arduino's digital pins. The relay contains an electromagnetic coil; when the control signal energises the coil, it generates a magnetic field that mechanically closes or opens the high-power circuit contacts, activating or deactivating the connected actuator. This arrangement ensures that the Arduino microcontroller is protected from electrical damage while maintaining full digital control over all actuators through the if-else control algorithm.

DHT22 Sensor and the 10 kΩ Pull-Up Resistor

The DHT22 is a capacitive humidity and temperature sensor that outputs a calibrated digital signal. Temperature measurement is based on a Negative Temperature Coefficient (NTC) thermistor embedded within the sensor, whose resistance decreases as temperature increases, following the Steinhart–Hart equation. Humidity measurement uses a capacitive element whose dielectric properties change with the concentration of water vapour between its plates. The sensor communicates with the Arduino via a single-wire serial protocol, requiring the data line to be held at a logic-high voltage when idle. To ensure this, a 10 kΩ pull-up resistor was connected between the data line and the 5V supply rail. Without this resistor, the data line floats at an undefined voltage, causing erratic or failed readings. The 10 kΩ value was selected as it provides sufficient current to maintain the logic-high state while limiting power dissipation, protecting the sensor from overcurrent damage. This is consistent with the standard wiring requirement specified in the DHT22 datasheet and applied in comparable low-cost environmental monitoring systems such as the nEMoS nano system reviewed in the portfolio.

L.O	Description	Concepts	Usage
CH.2.08	Galvanic and Electrolytic Cells	Conductive solutions	Understanding the basic idea behind capacitive deionization (CDI) which uses conductive solutions to operate
ME.2.05	Power	Define Power and its units of measure	Applying power and measuring help to calculate the output power of Pumps
MA.2.10	Data Analysis & Distribution	Percentiles z-scores Density curves	Analyzing sensors temperature, light intensity, humidity, and soil moisture data
PH.2.08	Electromagnet is Induction	Electromagnet induction	Need to understand how some sensors used to monitor environment for plants
MA.2.07	Derivatives & Rates of Change	Derivative Instantaneous /Average rate of Change	Modeling and analyzing the rate of change of parameter changes, or the rate of calibrating environmental conditions

Conclusions

In conclusion, the automated mini-greenhouse integrating a smart closed-loop environmental control system presents an effective and practical solution to Egypt's interlinked grand challenges of limited agricultural productivity, food insecurity, and environmental pollution. The system successfully maintained all three critical growth parameters for spinach cultivation within their predetermined optimal ranges: temperature between 10°C and 20°C, soil moisture between 60% and 80%, and light intensity between 15,000 and 25,000 lux, confirming that the prototype meets its core design requirements. The closed-loop control architecture, driven by an Arduino UNO and a combination of three sensor modules and three actuators, demonstrated reliable and responsive performance. Actuators corrected parameter deviations promptly upon sensor detection, confirming the effectiveness of the if-else control algorithm. The irrigation subsystem further demonstrated sustainability through its drainage and water-reuse mechanism, reducing resource consumption in line with the project's eco-friendly requirements. It was also concluded that the nEMoS monitoring approach is insufficient for applications requiring active environmental regulation, reinforcing the necessity of a full closed-loop architecture. The biological complexity of IMTA and the high infrastructure cost of IIWTTW further confirm that locally designed, affordable, sensor-based systems represent the most viable path forward for Egypt's agricultural context. The project demonstrates that an off-the-shelf component-based mini-greenhouse can enable consistent, out-of-season cultivation, simultaneously addressing four of Egypt's grand challenges.

Recommendation

Real-Life Application

The automated mini-greenhouse system has direct and scalable real-life applications across several sectors. In its most immediate form, the system is recommended for deployment at STEM High School as a hands-on learning platform that demonstrates embedded systems, closed-loop control, and environmental science in an integrated, practical setting. Students can use the mini-greenhouse to observe the measurable effects of temperature, light, and soil moisture on plant growth, reinforcing core STEM learning outcomes across chemistry, physics, and computer science. Beyond the school environment, the system can be scaled for community farming initiatives in urban areas and for commercial greenhouse operations on newly reclaimed desert land, supporting Egypt's agricultural expansion goals under Vision 2030. The addition of a Wi-Fi module to the microcontroller would further enable remote monitoring through a mobile application, enhancing its utility for larger deployments.

Enhancing Sensor Accuracy

The current prototype employs a DHT22 temperature sensor as noted in **figure(x)**, a resistive soil moisture sensor, and an LDR for light intensity measurement (Agus Siswoyo, March 2025). While these components provide reliable performance within the controlled prototype environment, they are subject to long-term drift in real outdoor deployments. It is recommended that future iterations replace the resistive soil moisture sensor with a capacitive sensor, which is significantly more resistant to corrosion and offers improved long-term stability. Substituting the LDR with a dedicated lux meter module such as the BH1750 would also provide more accurate and consistent light intensity readings, improving the precision of the light-regulating algorithm.

Energy Efficiency and Sustainability Upgrades

To further reduce operational costs and improve environmental sustainability, it is recommended to integrate a small solar panel array as a supplementary power source, as noted in **figure(x)**, paired with a LiFePO₄-based rechargeable battery backup (World Bank, 2025). This would make the mini-greenhouse fully off-grid capable, which is particularly valuable for deployment in rural or desert reclamation areas where grid electricity is unreliable. LiFePO₄ batteries offer high energy density, long cycle life, and excellent thermal stability, ensuring continuous system operation for powering sensors, grow lights, fans, and irrigation pumps reliably over extended periods.

Expanding Crop Compatibility

While the current system is optimized for spinach, the closed-loop control architecture is inherently adaptable to other crops by updating the environmental setpoint database stored in the Arduino's memory (Fred Searle, December, 2023). Future development should include a user-selectable crop profile interface, allowing operators to choose from a predefined library of crop-specific conditions such as peanuts, green beans, or tomatoes without requiring hardware modifications, noted in **figure(x)**. This enhancement would significantly expand the system's agricultural impact and make it a versatile tool for year-round cultivation of diverse crops across different Egyptian governorates.

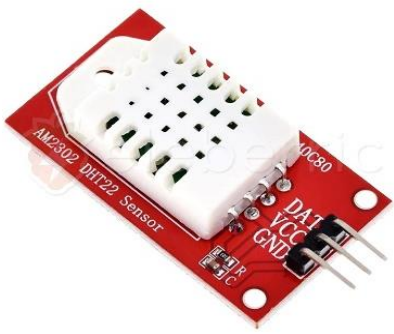


Figure 8: enhanced version of DHT22

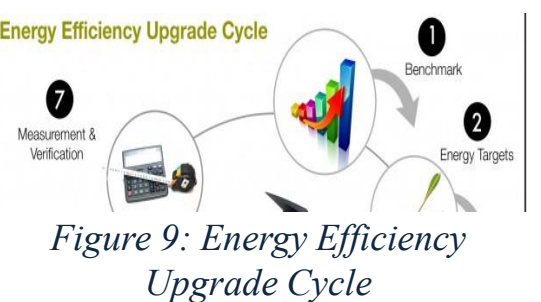


Figure 9: Energy Efficiency Upgrade Cycle



Figure 10: Expanding crop tool (HarvestEye)

Literature Cited

- 1915.82 - Lighting. | Occupational Safety and Health Administration. (n.d.). <https://www.osha.gov/laws-regs/regulations/standardnumber/1915/1915.82>
- Abbie. (2013, December 8). Do I have to connect a resistor to my DHT22 humidity sensor? Raspberry Pi Stack Exchange. <https://raspberrypi.stackexchange.com/questions/12161/do-i-have-to-connect-a-resistor-to-my-dht22-humidity-sensor>
- Ford, H. (2015). Demonstrating Osmotic Potential: One Factor in the Plant Water Potential Equation. Proceedings of the Association for Biology Laboratory Education, 36.
- MATLAB. (2018). Use of Peltier device as thermoelectric cooler. In MATLAB. Retrieved February 2, 2019, from <https://www.mathworks.com/help/spuug/use-of-peltier-device-as-thermoelectric-cooler.html>
- Zhang, Y. (n.d.). Design of a Closed-Loop Error-in-Variable System Controller and its application in Quadrotor UAV. IGI Global, 14(1), 23. <https://doi.org/10.4018/IJSSMET.321658>
- Mph, Z. S. (2026, March 4). What is thermoregulation, and how does it work? <https://www.medicalnewstoday.com/articles/thermoregulation>

Acknowledgment

First, we want to thank Allah for blessing us. Second, we would like to thank our teachers who helped us in many things and who played a great role in the completion of this efficient project, Mr.Mohamed Bekheet, the capstone General, Ms. Ninette R, our capstone leader, and Mr.Ahmed Hassan, our capstone teacher. Additionally, we would like to thank National Research Institute for counseling during the project.